تمرين 1

اكتب برنامج يعمل Array فيها **5 أرقام** ويطبعهم باستخدام for.

مثال الناتج:

```
10
20
30
40
50
```

Main.cs

```
// Online C# Editor for free
// Write, Edit and Run your C# code using C# Online Compiler

using System;

public class HelloWorld
{
    public static void Main(string[] args)
    {

        int[] num = new int [5];
        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32( Console.ReadLine());
        }
         Console.WriteLine("----------------------------------");
       foreach (int i in num)
       {
             Console.WriteLine(i);
       }
    }
}
```

Output

```
Enter any number
10
Enter any number
20
Enter any number
30
Enter any number
40
Enter any number
50
----------------------------------
10
20
30
40
50

=== Code Execution Successful ===
```

تمرين 2

اكتب برنامج يعمل Array من **5 أرقام** ويطبعهم باستخدام foreach.

Main.cs

```
// Online C# Editor for free
// Write, Edit and Run your C# code using C# Online Compiler

using System;

public class HelloWorld
{
    public static void Main(string[] args)
    {

        int[] num = new int [5];
        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32( Console.ReadLine());
        }
         Console.WriteLine("----------------------------------");
       foreach (int i in num)
       {
            Console.WriteLine(i);
       }
    }
}
```

Output

```
Enter any number
11
Enter any number
12
Enter any number
21
Enter any number
36
Enter any number
25
----------------------------------
11
12
21
36
25

=== Code Execution Successful ===
```

تمرين 3

اكتب برنامج يطلب من المستخدم إدخال **5 أرقام** ويخزنهم في Array ثم يطبعهم.

مثال:

```
Enter number: 10
Enter number: 20
Enter number: 30
Enter number: 40
Enter number: 50
```

الناتج:

```
10
20
30
40
50
```

Main.cs

```
// Online C# Editor for free
// Write, Edit and Run your C# code using C# Online Compiler

using System;

public class HelloWorld
{
    public static void Main(string[] args)
    {

      int[] num = new int [5];
      for (int i = 0; i < num.Length; i++)
      {
          Console.WriteLine("Enter any number");
          num[i] = Convert.ToInt32( Console.ReadLine());
      }
       Console.WriteLine("----------------------------------");


     foreach (int i in num)
     {
            Console.WriteLine(i);
     }
    }
}
```

Share Run

Output

```
Enter any number

Enter any number
2
Enter any number
3
Enter any number
4
Enter any number

----------------------------------
1
2
3
4
5

=== Code Execution Successful ===
```

تمرين 4

اكتب برنامج يطلب من المستخدم إدخال **5 أرقام** ثم يحسب **مجموعهم**.

Main.cs

```
// Online C# Editor for free
// Write, Edit and Run your C# code using C# Online Compiler

using System;

public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }
        Console.WriteLine("----------------------------------");

        int sum = 0;
        for (int i = 0; i < num.Length; i++)
        {
            sum += num[i];
        }
        Console.WriteLine("----------------------------------");
        Console.WriteLine("sum is: " + sum);
    }
}
```

Output

```
Enter any number
12
Enter any number
144
Enter any number
58
Enter any number
23
Enter any number

----------------------------------
----------------------------------
sum is: 238

=== Code Execution Successful ===
```

تمرين 5

اكتب برنامج يطلب من المستخدم إدخال **5 أرقام** ثم يطبع **أكبر رقم**.

مثال:

```
Max = 90
```

Main.cs

```
// Online C# Editor for free
// Write, Edit and Run your C# code using C# Online Compiler

using System;

public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");


        int max = num[0];

        for (int i = 1; i < num.Length; i++)
        {
            if (num[i] > max)
            {
                max = num[i];
            }
        }

        Console.WriteLine("----------------------------------");
        Console.WriteLine("max number is: " + max);
    }
}
```

Output

```
Enter any number
10
Enter any number
25
Enter any number
36
Enter any number
987
Enter any number
899
----------------------------------
----------------------------------
max number is: 987

=== Code Execution Successful ===
```

تمرين 6

اكتب برنامج يطلب من المستخدم إدخال **5 أرقام** ثم يطبع **أصغر رقم**.

مثال:

```
Min = 3
```

Main.cs

```
using System;
using System.Linq;

public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");


        int min = num.Min();
        Console.WriteLine("Min number is: " + min);
    }
```

Share Run

Output

```
Enter any number
12
Enter any number
25
Enter any number
6
Enter any number
5
Enter any number
8
----------------------------------
Min number is: 5

=== Code Execution Successful ===
```

تمرين 7

اكتب برنامج يطلب من المستخدم إدخال **10 أرقام** ثم يطبع **الأرقام الزوجية فقط**.

مثال:

```
2
4
6
8
```

Even:

Main.cs

```
using System;
using System.Linq;

public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");

       Console.WriteLine("Even numbers are:");

        for (int i = 0; i < num.Length; i++)
        {
            if (num[i] % 2 == 0)
            {
                Console.WriteLine(num[i]);
            }
        }
    }
}
```

Output

```
Enter any number
22
Enter any number
45
Enter any number
68
Enter any number
11
Enter any number
8
----------------------------------
Even numbers are:
22
68
8

=== Code Execution Successful ===
```

Odd:

Main.cs

```
using System;
using System.Linq;

public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");

       Console.WriteLine("odd numbers are:");

        for (int i = 0; i < num.Length; i++)
        {
            if (num[i] % 2 != 0)
            {
                Console.WriteLine(num[i]);
            }
        }
    }
}
```

Output

```
Enter any number
2
Enter any number
55
Enter any number
13
Enter any number
8
Enter any number
10
----------------------------------
odd numbers are:
55
13

=== Code Execution Successful ===
```

تمرين 8

اكتب برنامج يطلب من المستخدم إدخال **5 أرقام** ثم يحسب **متوسطهم (Average)**.

مثال:

Average = 25

Main.cs

Share Run

```
using System;
using System.Linq;

public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");



        double avg = num.Average();
        Console.WriteLine("Average is: " + avg);
    }
}
```

Output

```
Enter any number
25
Enter any number
14
Enter any number
45
Enter any number
85
Enter any number
79
----------------------------------
Average is: 49.6

=== Code Execution Successful ===
```

تمرين 9

اكتب برنامج يعمل **Array فيها 5 أرقام** ثم يعكس ترتيبها.

مثال:

Array:

1 2 3 4 5

Main.cs

Share Run

```
using System;
using System.Linq;

public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");

        int[] reversed = num.Reverse().ToArray();

        Console.WriteLine("Reversed numbers:");

        foreach (int i in reversed)
        {
            Console.WriteLine(i);
        }
    }
}
```

Output

```
Enter any number
25
Enter any number
87
Enter any number
98
Enter any number
21
Enter any number
6
----------------------------------
Reversed numbers:
6
21
98
87
25

=== Code Execution Successful ===
```

تمرين 10

اكتب برنامج يبحث عن رقم داخل Array.

لو الرقم موجود يطبع:

```
Number found at index 2
```

ولو مش موجود:

```
Number not found
```

اذا الرقم موجود:

Main.cs

```
public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");

        Console.WriteLine("Enter number to search:");
        int search = Convert.ToInt32(Console.ReadLine());

        bool found = false;

        for (int i = 0; i < num.Length; i++)
        {
            if (num[i] == search)
            {
                Console.WriteLine("Number found at index " + i);
                found = true;
                break;
            }
        }

        if (!found)
        {
            Console.WriteLine("Number not found");
        }
    }
}
```

Output

```
Enter any number
12
Enter any number
25
Enter any number
69
Enter any number
47
Enter any number
65
----------------------------------
Enter number to search:
12
Number found at index 0

=== Code Execution Successful ===
```

اذا الرقم ما موجود:

Main.cs

```
public class HelloWorld
{
    public static void Main(string[] args)
    {
        int[] num = new int[5];

        for (int i = 0; i < num.Length; i++)
        {
            Console.WriteLine("Enter any number");
            num[i] = Convert.ToInt32(Console.ReadLine());
        }

        Console.WriteLine("----------------------------------");

        Console.WriteLine("Enter number to search:");
        int search = Convert.ToInt32(Console.ReadLine());

        bool found = false;

        for (int i = 0; i < num.Length; i++)
        {
            if (num[i] == search)
            {
                Console.WriteLine("Number found at index " + i);
                found = true;
                break;
            }
        }

        if (!found)
        {
            Console.WriteLine("Number not found");
        }
    }
}
```

Output

```
Enter any number

Enter any number
14
Enter any number
55
Enter any number
89
Enter any number
90
----------------------------------
Enter number to search:
91
Number not found

=== Code Execution Successful ===
```